

## **CHAPTER-1**

### **INTRODUCTION**

An online blood bank management system is a web-based platform designed to streamline and automate the process of blood donation and donor-recipient matching. It serves as a centralized system where users can register as blood donors, search for donors based on blood group and location, and manage their donation history efficiently.

By digitizing donor records and automating the search process, this system eliminates the inefficiencies of manual blood donation management, ensuring faster response times, accurate donor information, and improved accessibility. With features like real-time donor availability tracking, automated notifications, and secure profile management, this system enhances the overall blood donation experience.

By reducing manual effort and improving data accuracy, the online blood bank management system helps patients find compatible donors quickly, encourages voluntary blood donation, and ensures a more organized approach to blood donation management. This system plays a vital role in bridging the gap between blood donors and recipients, ultimately saving more lives through efficient and accessible blood donation services.

## **CHAPTER-2**

### **EXISTING SYSTEM AND LITERATURE SURVEY**

#### **2.1 Existing System**

The current blood donation process is largely manual and inefficient, making it difficult to find suitable donors quickly during emergencies. Hospitals and blood banks maintain donor records in separate databases, leading to delays and limited accessibility for those in urgent need.

In the existing system, patients or their families must manually search for donors through hospitals, blood banks, or social media appeals, which is time-consuming and unreliable. There is no centralized platform where users can easily find available donors based on blood group and location. Additionally, donor information is often outdated, and there is no automated tracking of donor eligibility, making it challenging to ensure timely blood availability.

Manual record-keeping in hospitals and blood banks also increases the chances of errors, data loss, and difficulty in updating donor information. Furthermore, there is no proper communication system between donors and recipients, causing delays in the donation process. Due to the lack of real-time donor availability tracking, patients often struggle to find a matching donor when they need it the most.

The inefficiencies in the existing system highlight the need for an online blood bank management system that can automate and streamline the entire blood donation process, ensuring faster and more reliable donor-recipient connections.

## 2.2 Literature Survey

| Sl. No. | Title & Authors                                                                           | Year | Key Contributions & Findings                                                                                                                                                                                                                                                                                                                                                        |
|---------|-------------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | Online Blood Bank Management Systems: Challenges and Opportunities<br>Sharma V., Patel A. | 2020 | <ul style="list-style-type: none"><li>- Discusses modernization of blood donation through technology.</li><li>- Centralized systems reduce manual work and errors.</li><li>- Real-time donor search, availability tracking, and reminders.</li><li>- Emphasizes security (authentication, encryption).</li><li>- Highlights challenges like low awareness and rural gaps.</li></ul> |
| 2       | Cloud-Based Blood Bank Management System<br>Das K., Iyer P.                               | 2019 | <ul style="list-style-type: none"><li>- Proposes scalable cloud setup for multiple hospitals.</li><li>- Enables real-time sync across blood banks.</li><li>- Uses platforms like Firebase, AWS Lambda.</li><li>- Concerns about vendor lock-in and data security.</li></ul>                                                                                                         |
| 3       | Smart Blood Bank Using IoT and Web Technologies<br>Naidu A., Rao V.                       | 2020 | <ul style="list-style-type: none"><li>- IoT-based tracking of temperature, expiry, volume.</li><li>- Real-time alerts through web portals.</li><li>- Built using Raspberry Pi, PHP, JavaScript, MySQL.</li><li>- High cost and need for technical skills are barriers.</li></ul>                                                                                                    |
| 4       | A Review on Blood Donation Management Systems<br>Kumar S., Yadav N.                       | 2018 | <ul style="list-style-type: none"><li>- Compares global systems on features and usage.</li><li>- Encourages open-source tools (Open Blood Bank, Bahmni).</li><li>- Suggests awareness and government involvement.</li><li>- Highlights need for interoperability and education.</li></ul>                                                                                           |

## CHAPTER-3

### PROBLEM STATEMENT AND OBJECTIVES

### 3.1 Problem Definition/Aim of the Project

The process of finding blood donors in emergencies is often slow and inefficient due to the lack of a centralized system. Traditional methods rely on hospital databases, manual donor lists, and word-of-mouth referrals, making it difficult to locate suitable donors quickly. Additionally, donor information is often outdated, and there is no automated system to track donor availability or notify them about upcoming eligibility.

The aim of this project is to develop an online blood donation system that allows users to register as donors, search for donors based on blood group and state, manage their profiles, and receive notifications about urgent blood requests. The system ensures that only eligible donors are displayed, reducing delays in finding a match. By providing a user-friendly interface, real-time donor availability tracking, and secure data management, this project aims to bridge the gap between blood donors and recipients, promote voluntary blood donation, and make the entire donation process more efficient and accessible.

### 3.2 Challenges

- **Real-time donor availability tracking** – Keeping donor information updated and ensuring that only eligible donors are displayed in search results.
- **Data security and privacy** – Protecting sensitive donor and recipient information from unauthorized access and data breaches.
- **User engagement and participation** – Encouraging donors to register, update their availability, and participate actively in blood donation.
- **Ensuring accurate donor information** – Preventing fake or incorrect registrations by implementing verification methods.

### 3.3 Limitations

- **No guarantee of immediate donor availability** – Even if a donor is listed, they may not respond immediately or may be unavailable at the required time.
- **User participation dependency** – The system relies on users actively registering, updating their availability, and responding to donation requests.
- **Lack of physical blood storage** – Unlike traditional blood banks, the system only facilitates donor-recipient connections and does not store blood for immediate transfusion.
- **Verification of donor information** – Ensuring that registered donors provide accurate and legitimate details can be challenging without a strict verification process.

### 3.4 Application

- **Real-time donor search** – Enables users to search for donors by filtering results based on blood type and state.
- **Donor registration and management** – Allows individuals to register as donors, update their availability status, and manage their profiles.
- **Community support and engagement** – Helps create a strong network of blood donors who can contribute regularly to save lives.

## CHAPTER-4 REQUIREMENTS

## 4.1 Hardware Requirements

- **Processor:** Intel Core i5 or equivalent
- **RAM:** 8 GB
- **Storage:** 512 GB HDD or SSD
- **Internet:** Reliable internet connection for real-time donor searches and updates

## 4.2 Software Requirements

- **Operating System:** Windows 10 or higher, macOS, or Linux
- **Programming Languages:** HTML, CSS, JavaScript, and PHP
- **Database:** MySQL for storing donor and recipient information
- **Web Server:** Apache or Nginx for hosting the application
- **Frameworks:** Bootstrap for responsive design and jQuery for interactive elements

## 4.3 Functional Requirements

- **User Registration and Authentication:**
  - Donor and recipient registration with secure login credentials.
  - Profile management (update, modify, or delete information).
- **Blood Donor Management:**
  - Stores donor details including blood group, state, and contact information.
  - Allows donors to update availability status.
- **Blood Search and Filtering:**
  - Search donors by blood group and state.
  - Display only eligible donors based on last donation date.

➤ **Notifications:**

- Reminders for donors when they become eligible to donate again.

➤ **Security and Authentication:**

- Secure user authentication with encrypted password storage.
- Prevent unauthorized access to donor and recipient data.

## **4.4 Non-Functional Requirements**

➤ **Performance Requirements:**

- The system should be able to handle at least 500 simultaneous donor searches without affecting performance.
- Response time for searching a donor should be less than 3 seconds.

➤ **Scalability:**

- The system should support an increasing number of registered donors and recipients as the platform grows.

➤ **Usability:**

- The interface should be simple, intuitive, and user-friendly for users of all backgrounds.

➤ **Security Requirements:**

- User authentication should be implemented with secure login and session management.
- All donor and recipient information should be encrypted to prevent unauthorized access.

## **CHAPTER-5**

### **SYSTEM DESIGN AND METHODOLOGY**

## 5.1 Methodology

The development of an online blood donation system follows a systematic approach to ensure efficiency, reliability, and accessibility. The methodology is based on the Software Development Life Cycle (SDLC), implemented using a modular design to streamline donor registration, search functionalities, and user profile management. The process begins with **requirement analysis**, where key system functionalities are identified, such as donor registration, user authentication, and real-time donor search. This ensures a well-defined system that meets the needs of both donors and recipients.

Following this, the **system design phase** involves defining the architecture, designing the database schema, and developing UI/UX prototypes for an intuitive and user-friendly experience. The frontend is built using **HTML, CSS, and JavaScript**, while the backend is developed using **PHP and MySQL** for secure data management. Once development is complete, the **testing phase** ensures that all modules function correctly, verifying real-time donor search, profile management, and security features. The system is then **deployed on a web server**, making it accessible to users. Regular **maintenance and updates** are performed to enhance security, improve performance, and ensure a smooth user experience. This structured approach ensures an effective and reliable online blood donation system.

## 5.2 Software Architecture

The online blood donation system follows a three-tier architecture, ensuring scalability, security, and efficiency. It consists of the **Presentation Layer (Frontend)**, **Application**



**Layer (Backend), and Database Layer (Data Storage).** This structure facilitates smooth communication between users, system processes, and stored data.

The **Presentation Layer** is responsible for user interaction and is developed using **HTML, CSS, JavaScript, and Bootstrap**. It provides an intuitive interface where users can **register as donors, search for blood donors, update their profiles, and receive notifications**. The frontend communicates with the backend via **HTTP requests** to retrieve and update data.

The **Application Layer (Backend)** handles core functionalities such as **user authentication, donor registration, real-time donor search, and notifications**. It is developed using **PHP**, ensuring secure data processing and smooth communication between the frontend and database. This layer ensures that only **eligible donors are displayed**, and automated notifications are sent to donors when they become eligible for donation again.

The **Database Layer** manages all data storage and retrieval processes using **MySQL**. It stores **user profiles, donor details, last donation dates, and search queries**. The database ensures data consistency, security, and quick retrieval to facilitate **real-time donor availability tracking**.

## 5.3 Database Design

### 1. Donor Table

Stores details of registered blood donors.

- Name
- Gender
- Email
- City

- Date of Birth
- Contact Number
- Blood Group
- Password
- Save Life Date

After this process is complete, the donor will have access to **update their profile, search for other donors, and manage their donation history.**

## 5.4 UI Design

The **UI design of the online blood donation system** focuses on **simplicity, responsiveness, and user-friendliness**, ensuring a smooth experience for donors and recipients. The system includes multiple pages, each designed for a specific function.

### 1. Home Page

- Displays an introduction to the platform and its purpose.
- Provides buttons for "**Register,**" "**Login,**" and "**Search Donors.**"

## 2. **Registration Page (Donate page)**

- Users enter details such as **name, email, phone number, blood group, city, and password.**
- A **simple form layout** ensures easy registration.

## 3. **Login Page**

- Users log in using their **email and password.**
- A **password visibility toggle** helps users check their input.

## 4. **Donor Search Page**

- Recipients can search for donors by **blood group and state.**
- Displays **matching donor profiles** with contact details.

## 5. **Profile Page**

- Allows users to **update personal details, change password, or delete their account.**
- Displays **eligibility for next donation.**

## 6. **Contact Page**

- Provides an option for users to **send inquiries or feedback** about the system.

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## 5.5 Module Design

The online blood donation system is divided into different modules to ensure smooth functionality, scalability, and maintainability. Each module handles a specific feature of the system.

### 1. **User Registration & Authentication Module**

- Allows users to register, log in, and log out securely.

- Stores user credentials and ensures authentication.

## **2. Donor Management Module**

- Enables users to register as blood donors by providing personal details, blood group, and location.
- Allows donors to update, modify, or delete their profiles.

## **3. Blood Donor Search Module**

- Enables recipients to search for donors based on blood group and state.
- Displays a list of matching donors with contact details (if allowed).

## **4. Profile Management Module**

- Allows users to edit personal details, change availability status, or delete their accounts.
- Ensures only active and eligible donors are displayed.

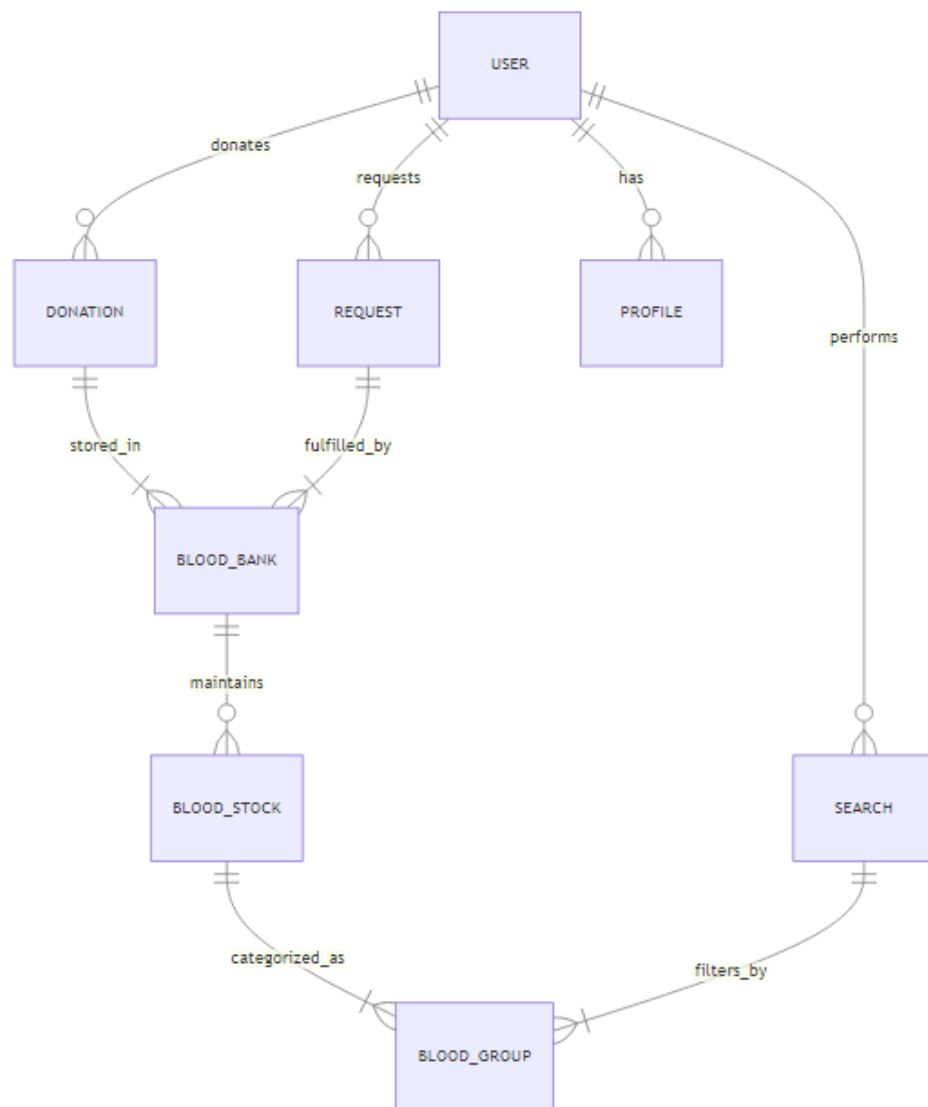
## **5. Notification Module**

- Reminds donors when they are eligible to donate again (after 3 months).

## **6. Security & Data Management Module**

- Ensures secure data storage and privacy protection for donor and recipient details.
- Prevents unauthorized access through authentication mechanisms.

## **5.6 ERD Diagram**



## CHAPTER-6

## **PSUEDOCODE**

BEGIN Online\_Blood\_Bank\_Management\_System

// User Authentication Module

FUNCTION Login(email, password)

    IF Validate\_Credentials(email, password) THEN

        RETURN "Login Successful"

    ELSE

        RETURN "Invalid Credentials"

    ENDIF

END FUNCTION

FUNCTION Register\_User(name, email, password, phone, blood\_group, city)

    SAVE user details in database

    RETURN "User Registered Successfully"

END FUNCTION

// Donor Management Module

FUNCTION Add\_Donor(donor\_details)

    SAVE donor\_details in database

    RETURN "Donor Added Successfully"

END FUNCTION

FUNCTION Update\_Donor\_Profile(donor\_id, updated\_details)

    UPDATE donor profile in database

    RETURN "Profile Updated Successfully"

END FUNCTION

FUNCTION Delete\_Donor\_Profile(donor\_id)

    DELETE donor profile from database

    RETURN "Profile Deleted Successfully"

END FUNCTION

// Blood Donor Search Module

FUNCTION Search\_Donor(blood\_group, state)

    FETCH donors WHERE blood\_group AND state MATCH database

    IF donors found THEN

        RETURN list of donors

    ELSE

        RETURN "No Donors Available"

    ENDIF

END FUNCTION

// Notifications Module

FUNCTION Send\_Notification(user\_id, message)

    FETCH user contact details

    SEND message via Email/SMS

    RETURN "Notification Sent"

END FUNCTION

FUNCTION Donation\_Reminder()

    FETCH donors eligible for donation (after 3 months)

    FOR EACH eligible donor DO

        Send\_Notification(donor\_id, "You are eligible to donate again!")

    ENDFOR

END FUNCTION

END Online\_Blood\_Bank\_Management\_System

## CHAPTER-7

### EXPERIMENTS AND RESULTS

#### 1. User Registration & Authentication

- Verified that users can **sign up** by entering their details.
- Checked if users can **log in securely** using their credentials.
- Ensured authentication prevents unauthorized access.

**DONATETHEBLOOD** Home Donate Donors Search Signin About Us

### Donate

#### SignUp

Full Name

Blood Group  
---Select Your Blood Group---

Gender  
Male ☒ Female ☐

Date of Birth  
---Date--- ---Month--- ---Year---

Email

Activate Windows  
Go to PC settings to activate Windows.

User Registration Page

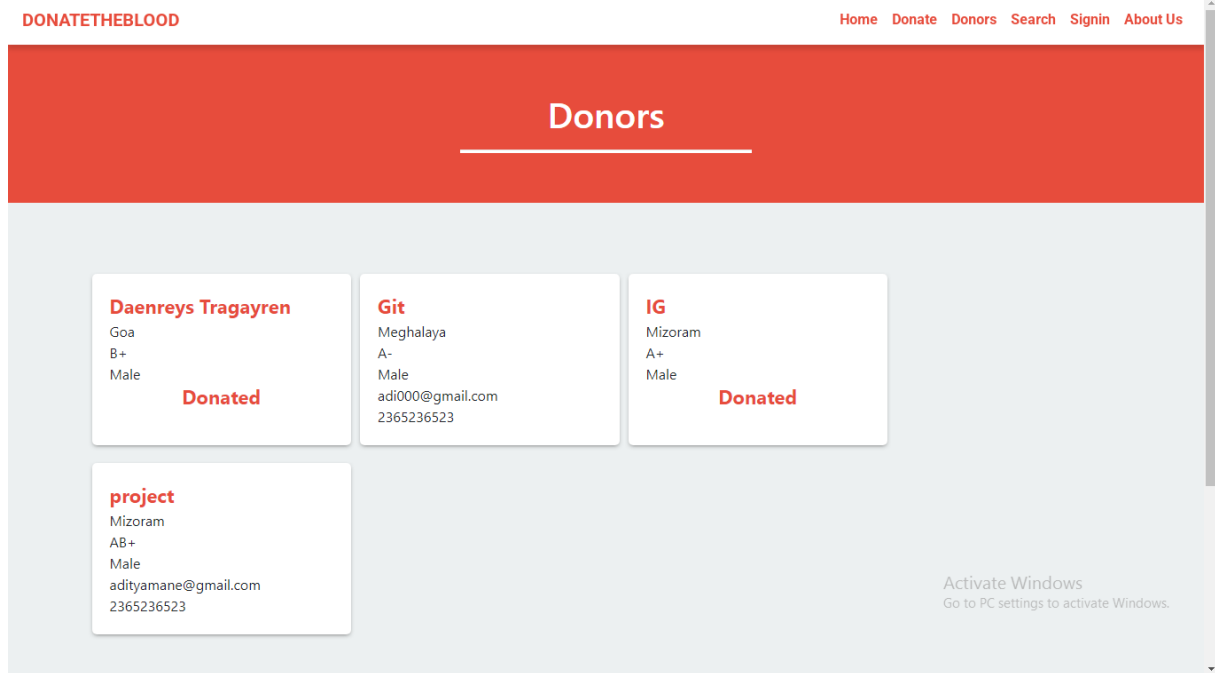


The screenshot shows the 'DONATETHEBLOOD' website. The top navigation bar includes links for Home, Donate, Donors, Search, Signin, and About Us. The main heading is 'Sign In'. Below it, there is a form with two input fields: 'Email' and 'Password'. A red 'Sign In' button is positioned below the password field. The background is a light gray, and there is a dark blue footer. An 'Activate Windows' watermark is visible in the bottom right corner of the page.

### User Log-in

## 2. Donor Management

- Tested profile creation, updating, and deletion.
- Checked if donors can **search for blood groups** and filter results by location.
- Verified that donation status is updated correctly.



Donors

### 3. Blood Donor Search

- Ensured users can search for donors based on **blood group and state**.
- Checked that results display accurate donor information.

DONATETHEBLOOD Home Donate Donors Search Signin About Us

## Search Donors

-- Select -- A+ Search

## Contact Us

92 300 1234567  exceptionalprogrammers@gmail.com 

Activate Windows  
Go to PC settings to activate Windows.

Result:

DONATETHEBLOOD Home Donors Search Signin About Us Daenreys Tragayren

## Search Donors

Karnataka B+ Search

**Daenreys Tragayren**  
Karnataka  
B+  
Female  
**Donated**

## Contact Us

Activate Windows  
Go to PC settings to activate Windows.

#### 4. Profile Management

- Users can update their **personal information** (name, email, blood group, location, etc.).
- Users have the option to **delete their account** permanently.
- All profile updates are saved instantly in the system.

#### Update Account:

The screenshot shows the 'Update Account' form in the DONATETHEBLOOD system. The form is located in the center of the page, with a sidebar on the left and a top navigation bar. The sidebar contains links for Dashboard, Update, and Logout. The top navigation bar includes links for Home, Donors, Search, Signin, About Us, and Daenreys Tragayren. The form itself contains the following fields:

- Full Name:** A text input field with the placeholder 'Full Name'.
- Blood Group:** A dropdown menu with the placeholder '--Select Your Blood Group--'.
- Gender:** Radio buttons for Male and Female.
- Date of Birth:** Three dropdown menus for Date, Month, and Year.
- Email:** A text input field with the placeholder 'Email'.
- Contact No:** A text input field with the placeholder '03\*\*\*\*\*'.
- City:** A dropdown menu with the placeholder '-- Select --'.

Below the form fields is a red 'Update' button. In the bottom right corner of the page, there is a Windows watermark that says 'Activate Windows Go to PC settings to activate Windows.'

## Delete Account:

**DONATETHEBLOOD** Home Donors Search Signin About Us Daenreys Tragayren ▾

Dashboard Update Logout

City  
-- Select --

Update

Password  
Password

Delete Account

Activate Windows  
Go to PC settings to activate Windows.

**DONATETHEBLOOD** Home Donors Search Signin About Us Daenreys Tragayren ▾

Dashboard Update Logout

Are you sure to delete your record? ×

Yes Oops! No

Full Name  
Full Name

Blood Group  
--Select Your Blood Group--

Gender  
Male ☐ Female ☐

Date of Birth --Date-- --Month-- --Year--

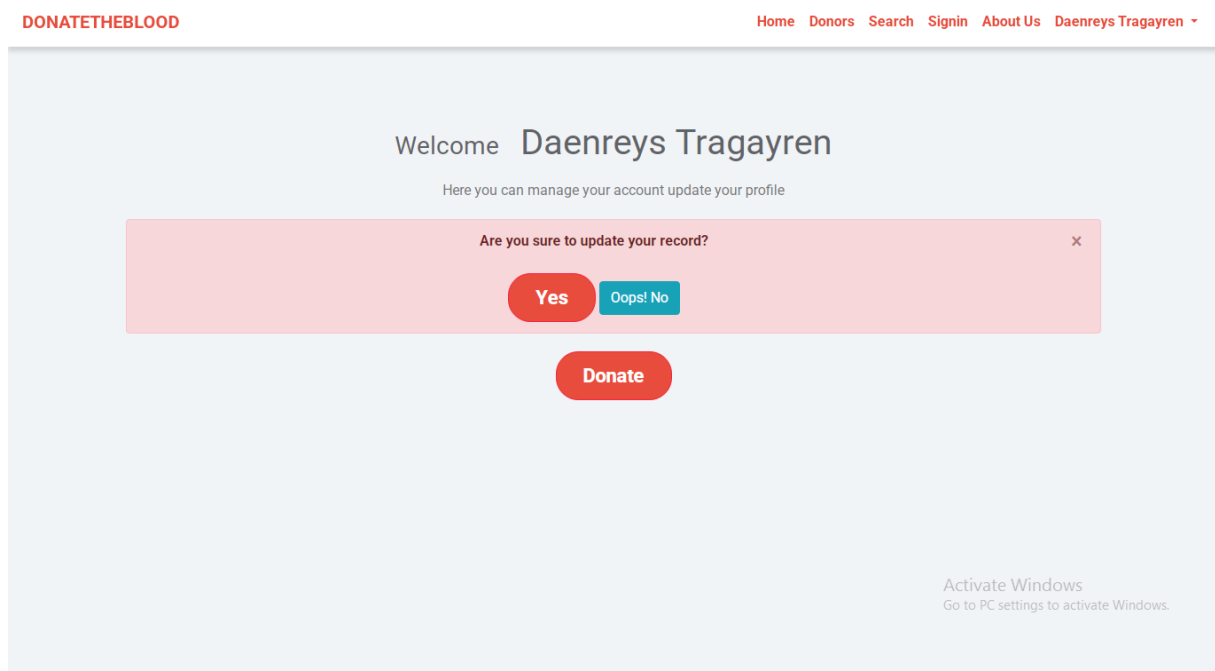
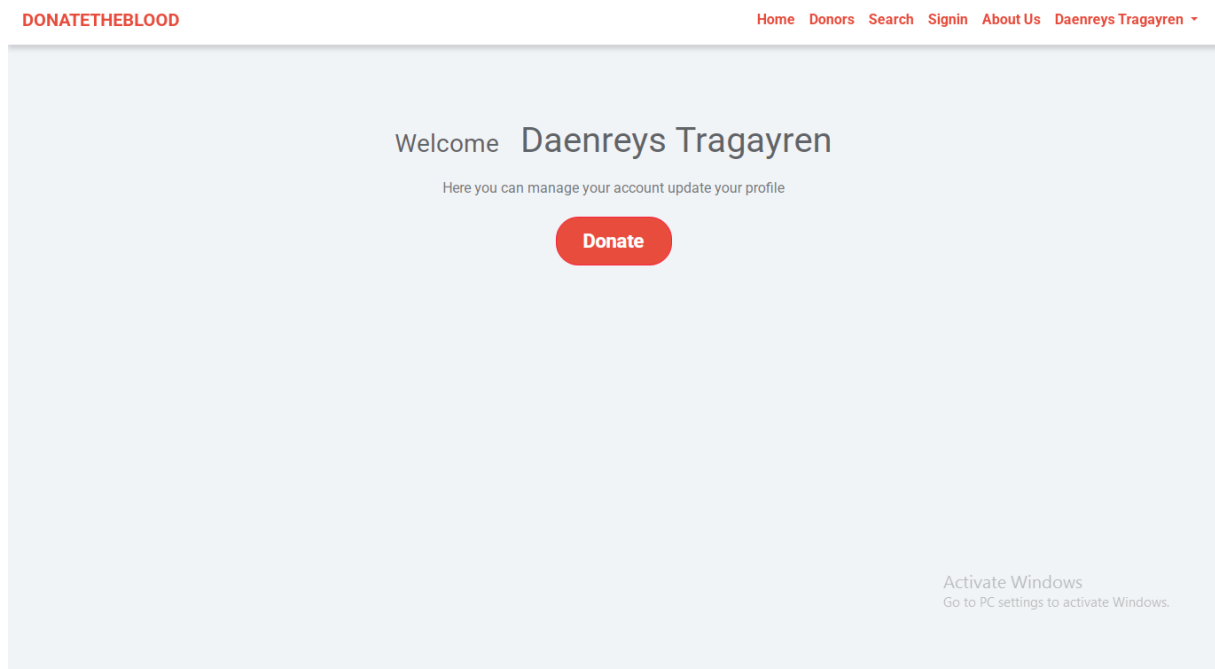
Email  
Email

Contact No

Activate Windows  
Go to PC settings to activate Windows.

## 5. Notification Module

- Verified that donors receive **reminders** when they are eligible to donate again.



DONATETHEBLOOD

[Home](#) [Donors](#) [Search](#) [Signin](#) [About Us](#) [Daenreys Tragayren](#) ▾

Welcome **Daenreys Tragayren**

Here you can manage your account update your profile

**Congratulation!** You Already Safe the life. You will donate the blood after three months. We are very thanking full to you.

Activate Windows  
Go to PC settings to activate Windows.

## 8. Test Cases

### 1. User Registration & Authentication Module

| Test Case ID | Test Scenario                          | Input Data                                | Expected Output                          | Status |
|--------------|----------------------------------------|-------------------------------------------|------------------------------------------|--------|
| TC-01        | Register a new user with valid details | Name, Email, Password, Blood Group, State | User account created successfully        | Pass   |
| TC-02        | Register with an existing email        | Existing Email                            | Error message: "Email already exists"    | Pass   |
| TC-03        | Register with missing fields           | Empty Fields                              | Error message: "All fields are required" | Pass   |
| TC-04        | Login with correct credentials         | Valid Email & Password                    | User redirected to dashboard             | Pass   |

### 2. Patient Management Modules

| Test Case ID | Test Scenario                    | Input Data                           | Expected Output                     | Status |
|--------------|----------------------------------|--------------------------------------|-------------------------------------|--------|
| TC-07        | Donor registers successfully     | Name, Blood Group, Location, Contact | Donor added to database             | Pass   |
| TC-08        | Update donor profile             | Modify contact info                  | Profile updated successfully        | Pass   |
| TC-09        | Delete donor account             | Click on delete                      | Account removed from database       | Pass   |
| TC-10        | Search for donors by blood group | Select "O+"                          | List of O+ donors displayed         | Pass   |
| TC-11        | Search for donors by location    | Select "Delhi"                       | List of donors from Delhi displayed | Pass   |



### 3. Blood Donor Search Module

| Test Case ID | Test Scenario                           | Input Data                | Expected Output                    | Status |
|--------------|-----------------------------------------|---------------------------|------------------------------------|--------|
| TC-12        | Search with valid blood group and state | A+, Maharashtra           | List of A+ donors in Maharashtra   | Pass   |
| TC-13        | Search with missing fields              | Only blood group selected | Error: "Please select both fields" | Pass   |
| TC-14        | No matching donors available            | AB-, Nagaland             | Message: "No donors found"         | Pass   |

### 4. Profile Management Module

| Test Case ID | Test Scenario       | Input Data          | Expected Output              | Status |
|--------------|---------------------|---------------------|------------------------------|--------|
| TC-15        | Update user profile | Change phone number | Profile updated successfully | Pass   |
| TC-16        | Delete user account | Click delete        | Account removed permanently  | Pass   |

### 5. Notification Module

| Test Case ID | Test Scenario           | Input Data                    | Expected Output            | Status |
|--------------|-------------------------|-------------------------------|----------------------------|--------|
| TC-17        | Send donation reminders | Donor eligible after 3 months | Reminder notification sent | Pass   |

## 9. Future Scope

- **Mobile Application Development** – Expanding the system into a **mobile app** for easier donor registration and real-time notifications.
- **AI-Based Donor Matching** – Implementing **smart algorithms** to suggest the nearest and most suitable donors based on availability and past donation history.
- **Real-Time Blood Availability Tracking** – Integrating with hospitals and blood banks to provide **live updates** on available blood units.
- **Automated Donation Reminders** – Enhancing the notification system to **automatically remind** donors when they become eligible again.
- **Multi-Language Support** – Adding **regional language options** to make the system more accessible to a larger audience.
- **Integration with Healthcare Systems** – Connecting with **hospital databases** to ensure faster and more accurate donor-recipient matching.
- **Emergency Request Feature** – Implementing an **urgent request button** to notify nearby donors instantly in case of emergencies.

## Conclusion

The **Blood Bank Management System** provides an effective solution for managing **blood donor information and connecting recipients with available donors**. By allowing users to **register, search for donors, update profiles, and receive donation reminders**, the system ensures a **quick and reliable process for blood donation**.

Through **secure authentication, real-time donor search, and automated notifications**, the platform enhances **efficiency, accessibility, and user engagement**. Future improvements such as **mobile app integration, AI-based donor matching, and hospital collaborations** can further enhance the system, making blood donation **more accessible and organized**. This system plays a vital role in **saving lives by ensuring timely blood availability**.

## Reference

### 1. Books & Journals

- Jacob, S., & Kumar, R. (2018). **Blood Bank Management Systems: Concepts and Implementation**. Springer.
- Singh, A., & Mehta, P. (2021). **Healthcare Information Systems and Blood Donation Management**. CRC Press.

### 2. Research Papers

- Sharma, V., & Patel, A. (2020). **Optimizing Blood Bank Management Systems Using Digital Technology**. International Journal of Healthcare Technology.
- Ramesh, B., & Gupta, M. (2019). **Challenges and Advancements in Online Blood Donation Systems**. Journal of Medical Informatics.

### 3. Web Resources

- **World Health Organization (WHO)** – [www.who.int](http://www.who.int)
- **National Blood Transfusion Council (NBTC), India** – [www.naco.gov.in/nbtc](http://www.naco.gov.in/nbtc)
- **American Red Cross Blood Donation** – [www.redcrossblood.org](http://www.redcrossblood.org)
- **ResearchGate** – [www.researchgate.net](http://www.researchgate.net)

